

ARTIFICIAL INTELLIGENCE — TECHNICAL PRESENTATION

Goal-Based Agent for Healthcare Diagnosis

How an AI agent sets a diagnostic goal, reads through patient data, and reasons its way to the best treatment path

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What We'll Cover

1

What is a goal-based agent?

The core idea, explained without jargon

2

Why diagnosis needs goal-based thinking

Where simple rule systems fall short

3

Inside the system

The architecture, step by step

4

Setting the diagnostic goals

What the agent is actually trying to achieve

5

Reading the patient data

The data pipeline behind every decision

6

Deciding & choosing a treatment path

With a real walkthrough example

What Is a Goal-Based Agent?

In plain words

A goal-based agent doesn't just react to what's in front of it. It keeps a goal in mind — like "find out what's wrong with this patient" — and chooses each action because it moves closer to that goal.

Think of a doctor deciding which test to order next — not from a fixed checklist, but by asking "will this get me closer to a confident diagnosis?"

Where it sits among AI agent types



Increasing ability to plan ahead →

A healthcare diagnosis agent needs exactly this level: enough foresight to weigh "what test should I order next" against a clear objective, without needing full utility-style cost-benefit maths for every decision.

Why Diagnosis Needs Goal-Based Thinking



A Fixed Rule-Based System

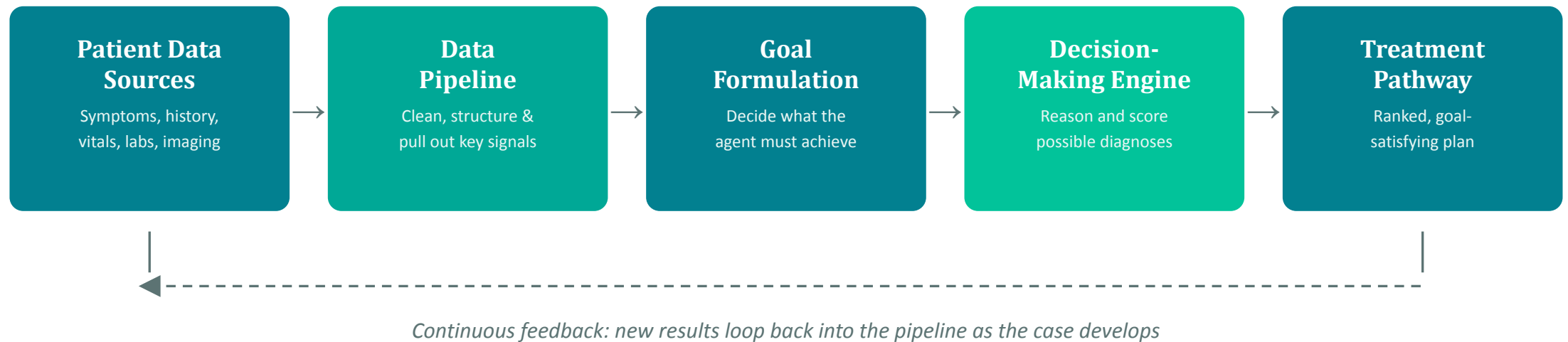
- Follows rigid if-this-then-that checklists
- Struggles the moment a case doesn't match a known pattern
- Can't explain “why” it asked for a particular test
- Treats every symptom combination as equally routine



A Goal-Based Agent

- Starts from the goal: “confirm or rule out condition X”
- Works backward to decide which data/tests actually matter
- Adapts its path when new results come in
- Weighs safety, accuracy, and speed together

System Architecture — The Big Picture



Each box is covered on its own slide next — starting with how the agent turns a patient's complaint into a concrete goal.

Step 1 — Setting the Diagnostic Goals

Before looking at any data, the agent turns the patient's complaint into a small set of concrete goals it must satisfy.

1

Primary Goal

Identify the single most likely condition explaining the symptoms.

2

Safety Goal

Rule out life-threatening causes first — e.g. check for a heart attack before assuming indigestion.

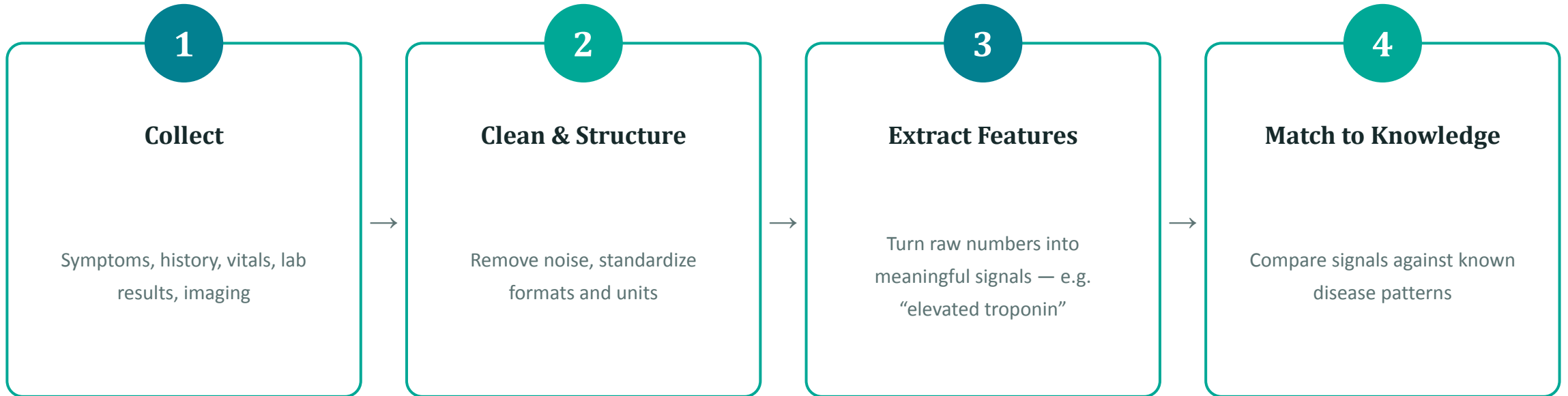
3

Efficiency Goal

Reach a confident answer using the fewest necessary tests — avoid over-testing.

Step 2 — Evaluating Patient Data

Raw patient information is messy. The pipeline turns it into something the agent can actually reason with.



Step 3 — Decision-Making Logic

The agent doesn't guess. It compares candidate diagnoses against its goals and scores each one — blending rule checks, probability, and machine-learning predictions.

Candidate Diagnosis	Matches Symptoms	Safety Risk if Wrong	Goal-Fit Score
Heart attack (cardiac event)	Moderate	Very high	9.2 / 10
Acid reflux / indigestion	Moderate	Low	5.4 / 10
Muscle strain	Low	Low	3.1 / 10

Even though “heart attack” isn't the most obviously matching symptom set, its high safety risk pushes its goal-fit score to the top — so the agent prioritizes ruling it out first.

In short: rule-based checks catch known danger signs, probability scoring weighs likelihood, and ML models spot subtle patterns — the agent combines all three against its goals.

Step 4 — Selecting the Optimal Treatment Pathway

Once a diagnosis satisfies the goals, the agent still has to choose how to act — weighing several real-world constraints at once.

Patient Constraints

Allergies, other conditions,
medications already in use

Urgency

How quickly action is needed
given the safety goal

Resources & Cost

What tests, staff, and
equipment are realistically
available

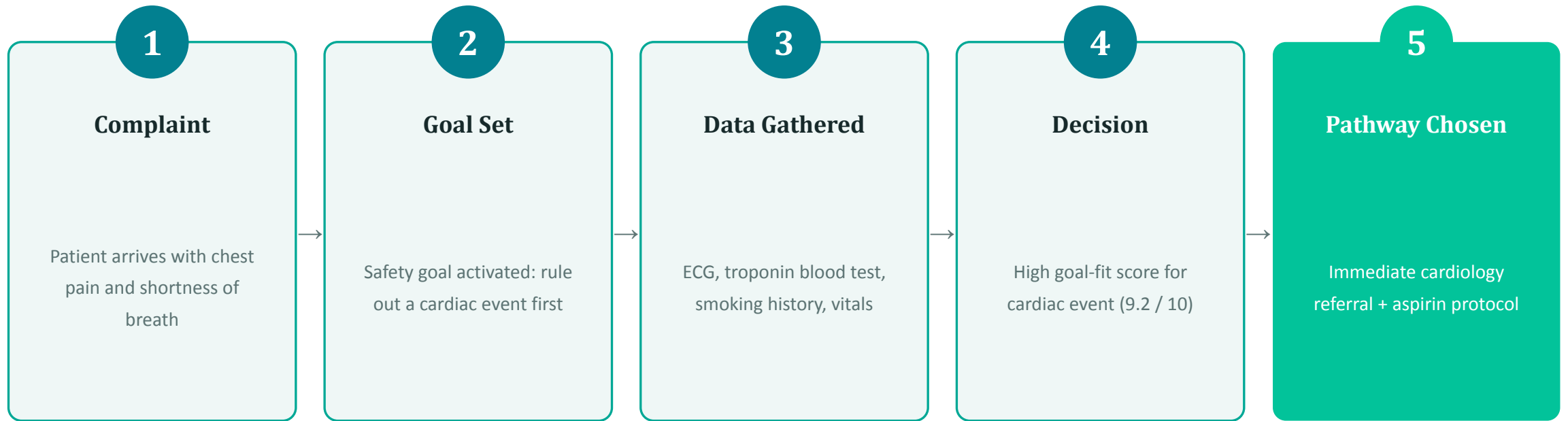
Clinical Guidelines

Standard protocols the plan
must stay consistent with

Output

A ranked treatment plan — not just one fixed answer — so the doctor can see the top option and the reasoning behind the alternatives.

Walkthrough: A Simple Example



The agent never treated “indigestion” as the default answer — the safety goal kept the dangerous possibility on top until it was properly ruled out.

Why This Matters

1

Faster, More Consistent Triage

The same goal-driven logic applies every time — no fatigue, no skipped steps.

2

Fewer Missed Critical Diagnoses

Safety goals keep dangerous possibilities in view until they're properly excluded.

3

Explains Its Reasoning

Goal-fit scores and matched signals give doctors a visible “why,” not a black box.

4

Supports Doctors, Doesn't Replace Them

Produces a ranked plan for a clinician to review and confirm.

CHALLENGES & CLOSING THOUGHTS

It's Not a Magic Fix

Data Quality & Bias

Poor or unrepresentative patient data leads to poor goal-fit scoring.

Explainability

Every recommendation must stay traceable back to real evidence.

Regulatory Approval

Medical AI systems face strict safety and validation requirements.

Workflow Integration

Must fit smoothly into how hospitals and clinicians already work.

Goal-based agents bring structured, transparent, goal-driven reasoning to a field where getting the diagnosis right matters most — as a partner to the clinician, not a replacement.

Thank You

Questions & Discussion